

A Face Biometric Module

Anti-Spoofing and Open-Set Recognition

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The plan

1. Problem & System
2. Two Detectors, and Fusion
3. Generalisation: the Domain Gap
4. From Liveness to Identity
5. Wrap-up

Problem & System

Why anti-spoofing?

A face recogniser is easily fooled by a **presentation attack**: a printed photo, a screen replay, a mask.

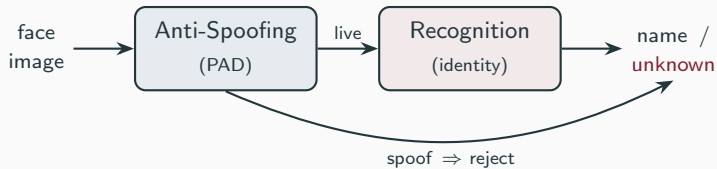
- Recognition alone answers “*who?*”
— but trusts the input is a real face.
- **Liveness** must come *first*: “*is this a live face or an attack?*”



live (bona fide) vs. spoof samples

A complete biometric system needs **both** stages.

The system at a glance



A cascade: a frame is matched against enrolled identities *only if* it passes the liveness check.

CelebA-Spoof (cropped) — one of the largest public anti-spoofing datasets.

- Large + public $\Rightarrow$ enough data to train a CNN.
- **Subject-disjoint test** split
 $\Rightarrow$ no identity leakage.
- Threshold fixed on *validation* only.

Split	Images	Spoof:Live
Train (fit)	67,306	$\sim 2:1$
Validation	11,877	$\sim 2:1$
Test	66,787	2.4:1

Imbalance handled with *class weights*.

Preprocessing & augmentation

Faces already cropped $\Rightarrow$ resize + normalise (ImageNet stats).

Augmentation **simulates the attack channel**:

- random JPEG, blur, color jitter
- flip, random crop, erasing

Goal: robustness to print/replay artefacts.



one image under random augmentations

Two Detectors, and Fusion

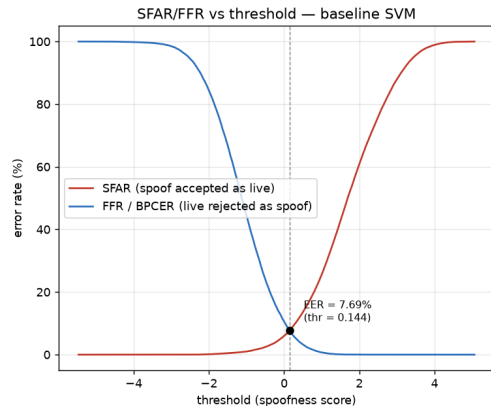
Approach 1 — classic baseline

Color-texture LBP + SVM

(Boulkenafet *et al.*)

- A photo of a face loses fine **texture**.
- Multi-scale LBP on YCbCr+HSV
⇒ **168** hand-crafted features.
- RBF-SVM, balanced classes.

A transparent *reference to beat*.

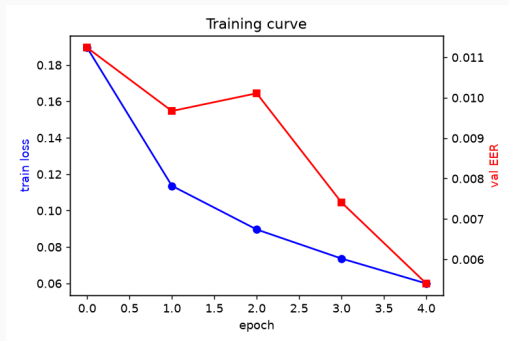


SFAR/FFR vs. threshold — EER = 7.69%

Approach 2 — fine-tuned CNN

ResNet-18, pre-trained on ImageNet,
fine-tuned end-to-end.

- Learns the features instead of hand-crafting them.
- AMP, class weights, cosine LR, early stopping on *val EER*.
- No overfitting: *val EER* $\rightarrow$ 0.5%.

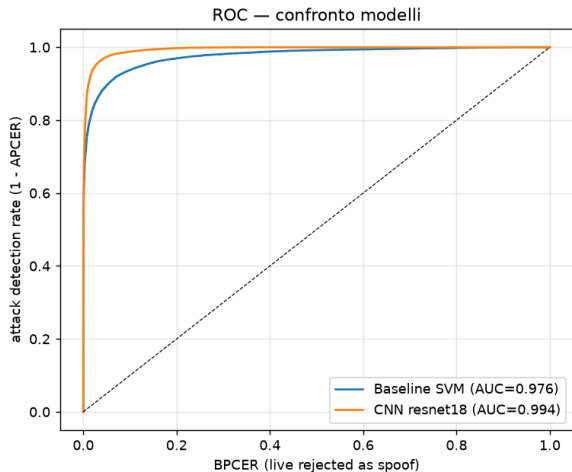


training / validation curves

Head-to-head: learned > hand-crafted

Metric	SVM	CNN
EER	7.69%	3.69%
AUC	0.976	0.994
BPCER@APCER=5%	12.8%	2.6%

The CNN **more than halves the EER** and dominates the ROC at every operating point.

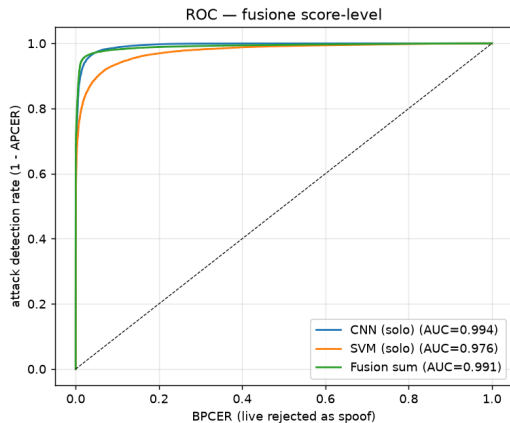


Score-level fusion

Two detectors fail on *different* images $\Rightarrow$ combine them at the score level.

- Scores must be normalised first.
- Min-max assumes known bounds — false for an unbounded SVM score.
- So compare it with **z-score** and **median/MAD**.

Norm. (test EER)	sum	weighted
min-max	3.35%	5.14%
z-score	3.82%	5.15%
median/MAD	3.65%	3.22%



Best overall: **median/MAD + weighted (3.22%)**. With the fragile min-max, the *tuned* weight overfits (5.14%).

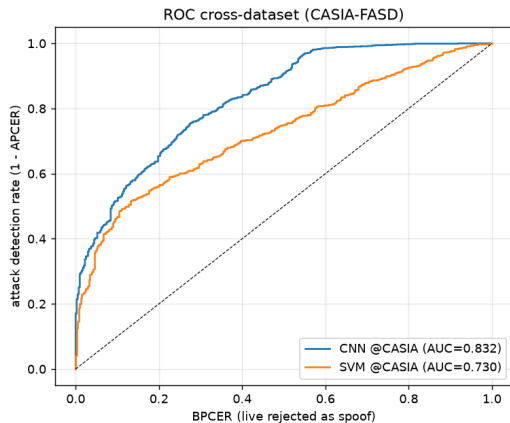
Generalisation: the Domain Gap

The honest result: cross-dataset

Same model, **unseen domain** (CASIA-FASD),
threshold fixed on the source.

Model	Scenario	HTER
CNN	intra (CelebA)	7.9%
CNN	cross (CASIA)	35.4%
SVM	intra (CelebA)	8.4%
SVM	cross (CASIA)	44.5%

A **threshold-transfer** failure, not lost power: the AUC stays at 0.83. The classic **domain gap** of PAD.



CNN still generalises *better* than the baseline (AUC 0.83 vs 0.73).

From Liveness to Identity

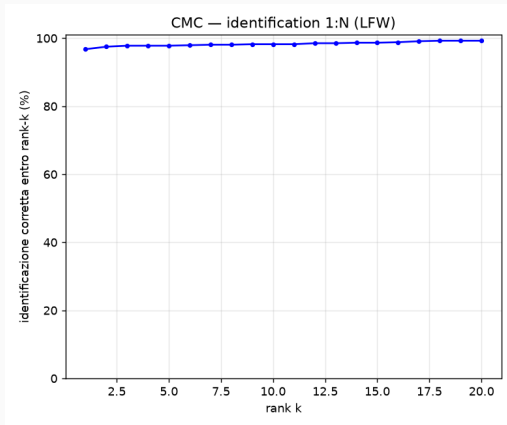
Adding recognition

A different problem $\Rightarrow$ a dedicated, pre-trained model:

- **MTCNN** detect + align.
- **FaceNet** (VGGFace2) $\rightarrow$ 512-d **embedding**.
- Compare by **cosine similarity**.

LFW benchmark

- Verification 1:1 — EER **3.30%**
- Identification 1:N — rank-1 **96.8%**



CMC curve on LFW

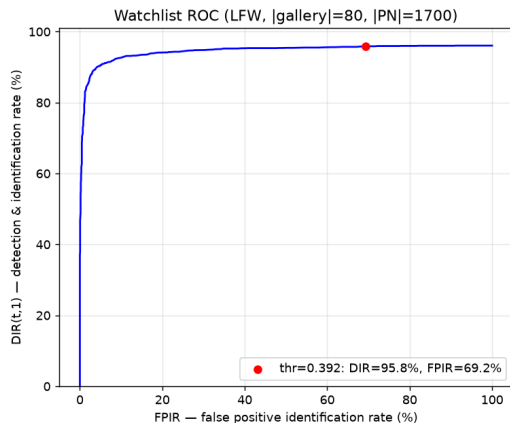
Open-set: a proper watchlist

Open-set: most people are *not* enrolled, so the system must answer “*unknown*”. Proper metrics: **DIR**, **FNIR**, **FPIR**.

LFW protocol: 80 enrolled, 2,544 genuine vs 1,700 impostor probes.

- Verification threshold (0.392)
does not transfer: FPIR = **69%**!
- Read the point off the ROC:
FPIR 1% → DIR 76.5%; FPIR 10% → DIR 92.7%.

The 2-subject personal gallery is only a *qualitative demo* (too small to measure).



same lesson as the domain gap: a threshold from one scenario is wrong in another.

Wrap-up

- Built a **full pipeline**: anti-spoof $\rightarrow$ open-set identity.
- **Learned** $>$ **hand-crafted**: the CNN more than halves the baseline EER (7.7% $\rightarrow$ 3.7%); robust fusion $\rightarrow$ 3.2%.
- Recognition benchmarked as a **proper watchlist** (DIR/FNIR/FPIR), not a binary claim.
- ★ The real lesson is **threshold transfer**: a point chosen in one setting is wrong in another (val $\rightarrow$ test, CelebA $\rightarrow$ CASIA, verification $\rightarrow$ watchlist) — measured honestly, not hidden.

Next: domain-generalisation training + lightweight on-site threshold calibration.

Thank you.

Questions?